

Catherine (Qi) Chen

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

GPA: 3.78 / 4.0

Aug 2025 – Present

Relevant Coursework: Introduction to Computer Systems, Principle of Functional Programming, Principles of Imperative Computation, Structure and Design of Digital Systems, Artificial Intelligence and Machine Learning, Computer Vision, Introduction to Electrical and Computer Engineering, Concepts of Mathematics, Matrix & Linear Transformation, Calculus in 3D

WORK EXPERIENCE

Shenzhen Pinghu Laboratory

Shenzhen, China

Software Developer

Jun 2025 – Aug 2025

- Developed a **document circulation web platform** that enable enterprises to directly obtain official documents from government authorities.
- Implemented a **multi-level approval workflow** with **6 approval stages** for applicant submission, supervisor review, and case-handler processing.
- Streamlined **government–enterprise communication** by enabling **real-time request tracking**, improving transparency, and significantly reducing approval turnaround time by 36%.
- Project received strong recognition from **senior leadership** for its practical impact.

Bohe Digital Technology

Shenzhen, China

Software & Hardware Developer

Sep 2023 – Aug 2024

- Contributed towards an **algorithmically optimized partition backlight**, enhancing display contrast for whiteboards and large outdoor displays; named **co-holder of the resulting patent**.
- Created **multiple mapping tables** depending on different specifications and configurations of display screens.
- Developed an **algorithm** to **detect and quantify** the brightness of each screen area for every frame, supported by an **external light sensor** connected to the internal backlights.

PROJECTS

Machine Learning Approach to Pathogen Distribution on Microplastics

Shenzhen, China

Researcher

Mar 2024 – Aug 2024

- Conducted research through the **Pioneer Data Science program** under Distinguished Professor **Sanjay Ranka** from the **University of Florida**.
- Published paper "**Machine Learning Approach to Predict Relative Abundance of Vibrio Spp. on Microplastics**" in **CPCI**.
- Collected **483 data points** on microplastics and physicochemical parameters of water body from the Bohai region in China.
- Developed an **artificial neural network model** in predicting pathogen spread via microplastics, achieving **87%** accuracy.

Abilitopia Team Matching Platform

Shenzhen, China

Full-Stack Developer

Jan 2023 – May 2024

- Established a website utilizing a **computational matching algorithm** to match students with teammates for class projects and extracurricular competitions based on **4 key criteria**: technical skill levels, grades, working dynamics, and MBTI profile.
- Generated **intelligent matching results** for like-minded students and provided **online learning resources** for remote areas.

Bioinformatics Analysis of Glucose Markers in Diabetes

Shenzhen, China

Researcher

Feb 2023 – Aug 2024

- Published paper "**Bioinformatics Analysis Identifies the Association Between Glucose Markers and Disease-Related Functions in Diabetes**" in the **International Journal of High School Research**, under the guidance of **Dr. Zhang Xingjian** from **Yale University**.
- Found diabetes datasets, plotted various **scatter plots and bar plots**, performed **linear regression, two sample t-tests** and **ANOVA** to identify significant relationships between related biomarkers.
- Invited to participate in **2 diabetes-related conferences** in Shenzhen, including the **2024 Shenzhen Diabetes Management Forum**, by **Dr. Lijing Jia**, Diabetes Committee Chairman of the Shenzhen Health Association.

SCIE Mathematics for Everyone Club

Shenzhen, China

Founder

Feb 2023 – Aug 2024

- Led curricular lessons in A-level coursework and taught **computational algorithms**.
- Designed **two Arduino devices**: one to assist dyslexia people and another to monitor study environments, showcased to **2000+ teachers and students** across campus.

SKILLS

Languages: C++11/14, C, Standard ML, Python3, Java, SystemVerilog

Web/Database: HTML, CSS, Vue 3.0, Element Plus, MySQL